Supplementary Information

Delta Dependency Prioritizes Paralog-Based
Synthetic Lethality Candidates Across Solid Tumor Types

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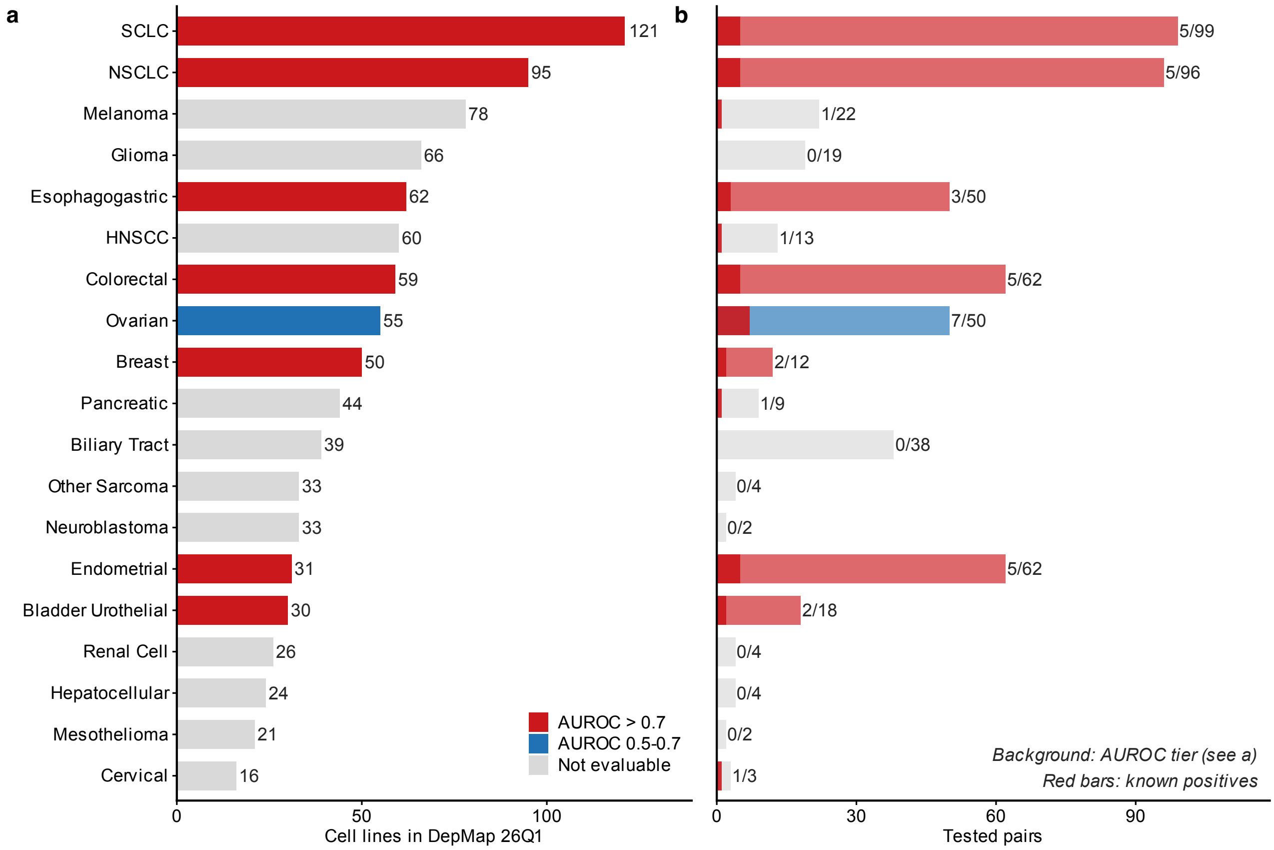

Fig. S1: Cell line landscape and evaluation summary across 23 solid tumor types. a, Cell line counts per cancer type, colored by DD AUROC tier (red: AUROC > 0.7; blue: 0.5–0.7; gray: < 0.5; light gray: not evaluable). b, Tested paralog pairs per cancer type; red bars show known gold-standard positives. Numbers indicate known/total pairs.

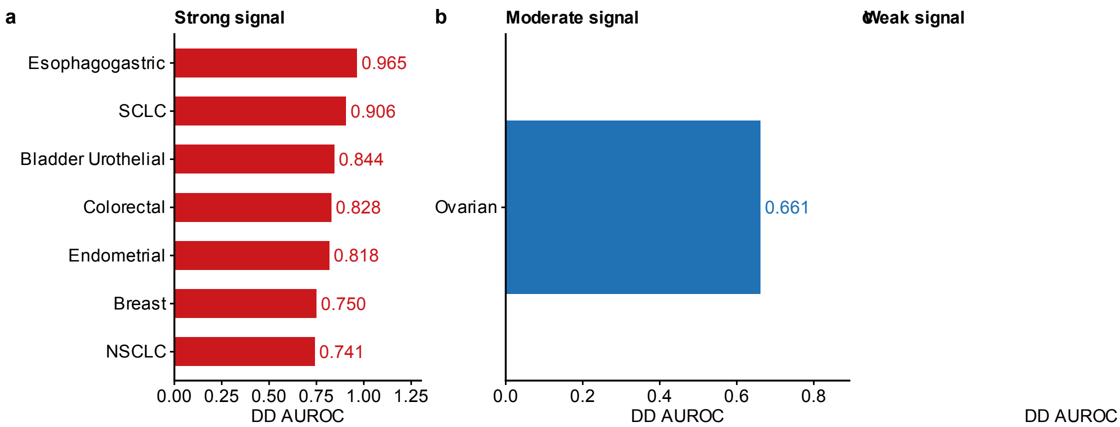

Fig. S2: Cross-cancer DD AUROC. DD AUROC for each of the 8 evaluable solid tumor types on the primary $\geq$5-per-group frame, grouped by signal strength. a, Strong signal (AUROC ≥ 0.7; 7 lineages: Esophagogastric 0.965, SCLC 0.906, Bladder Urothelial 0.844, Colorectal 0.828, Endometrial 0.818, Breast 0.750, NSCLC 0.741). b, Moderate signal (AUROC 0.5–0.7; Ovarian 0.661). Dashed line indicates random classifier (0.5). Bar labels show the number of tested paralog pairs per lineage. Cancer types with $<$2 known positive pairs were excluded from AUROC evaluation (see Table S1 for the 15 non-evaluable types). On the $\geq$3-per-group sensitivity frame, four additional lineages become evaluable (Biliary Tract 0.990, Pancreatic 0.949, Melanoma 0.617, Cervical 0.500; 9 of 12 evaluable lineages exceed 0.7).

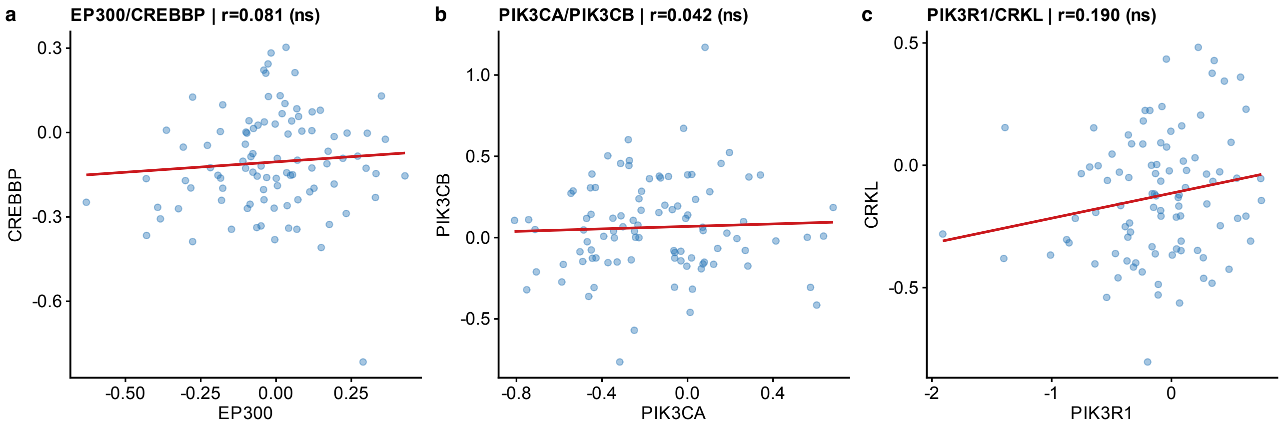

Fig. S3: CPTAC per-cohort scatter plots (UCEC). Representative scatter plot of EP300 vs. CREBBP protein abundance ($\log_2$ normalized intensity) in the UCEC cohort ($n = 83$). Each point represents one tumor sample. Line: linear regression fit with 95% confidence band. Pearson r and BH-corrected q-value shown. Similar scatter plots for other cohorts and paralog pairs are available upon request.

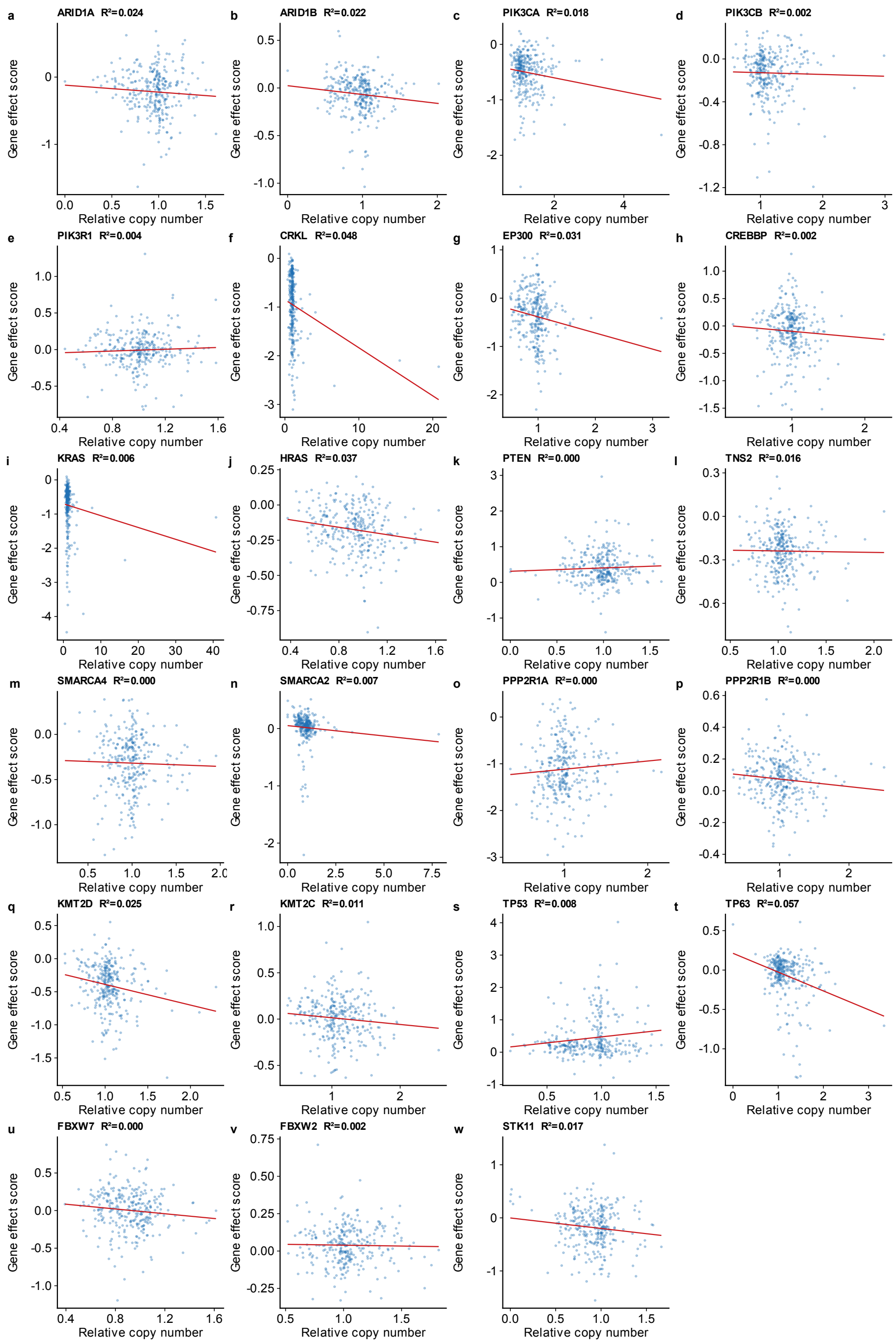

Fig. S4: CNV independence analysis. Scatter plots of relative copy number (x-axis) vs. Chronos gene-effect score (y-axis) for each of 23 paralog genes across 1,192 cell lines with both CNV and dependency data. Each panel shows one gene with the linear regression fit (red line) and R^2 value. All $R^2 < 0.10$, indicating that copy number variation explains <10% of gene-effect variance. Multivariate regression controlling for driver mutation and TP53 co-mutation left DD estimates essentially unchanged after CNV adjustment (see main text).

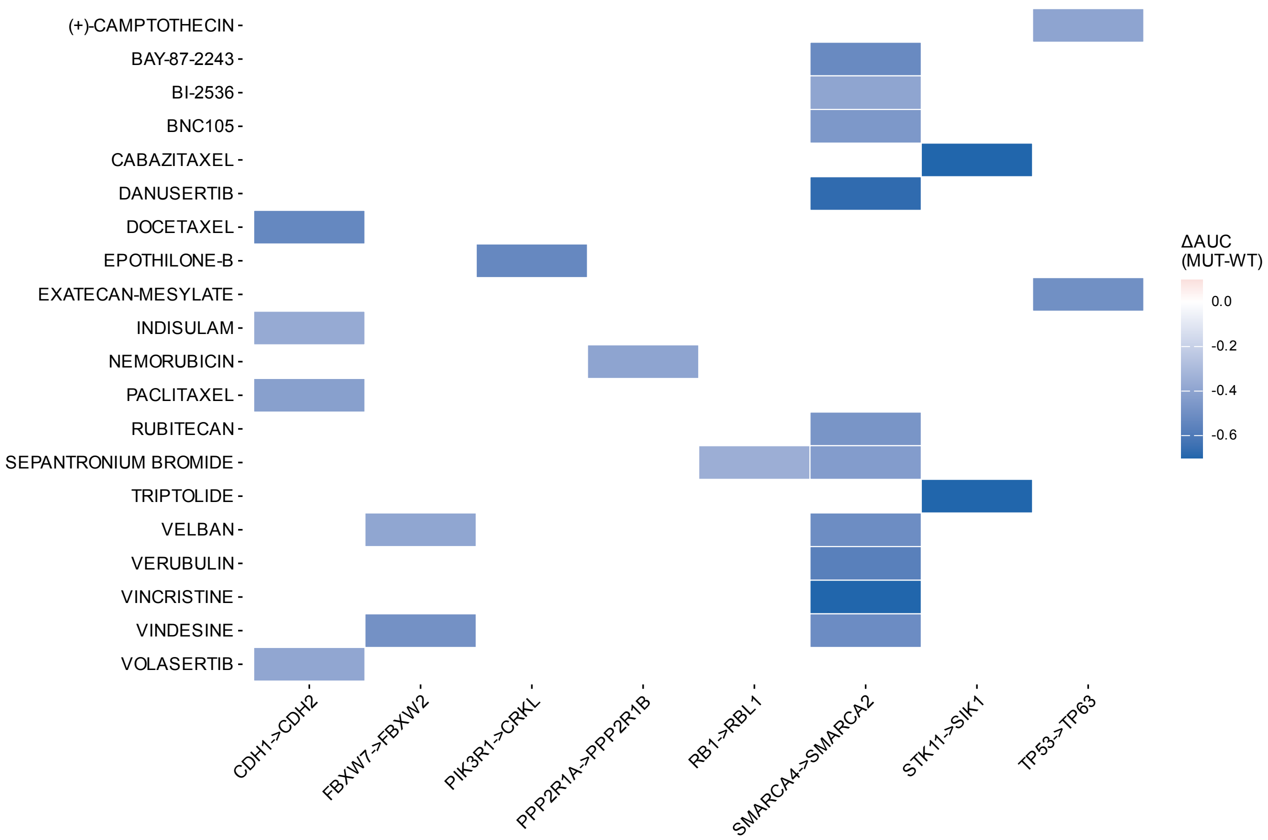

Fig. S5: PRISM drug selectivity heatmap. Heatmap of drug sensitivity differential (ΔAUC = mean AUC in driver-mutant minus mean AUC in wild-type cell lines) for the top compound–paralog pair associations from the PRISM Repurposing screen (1,482 compounds, 727 cell lines). Negative ΔAUC indicates selective sensitivity in driver-mutant cells. Associations shown met discovery-stage criteria (BH $q < 0.25$, $\Delta\text{AUC} < 0$, enrichment> 0.1). Known drug–target biology is recovered: MEK inhibitors with KRAS-mutant lines, mTOR/AKT inhibitors with PTEN-mutant lines, HDAC inhibitors with EP300-mutant lines.

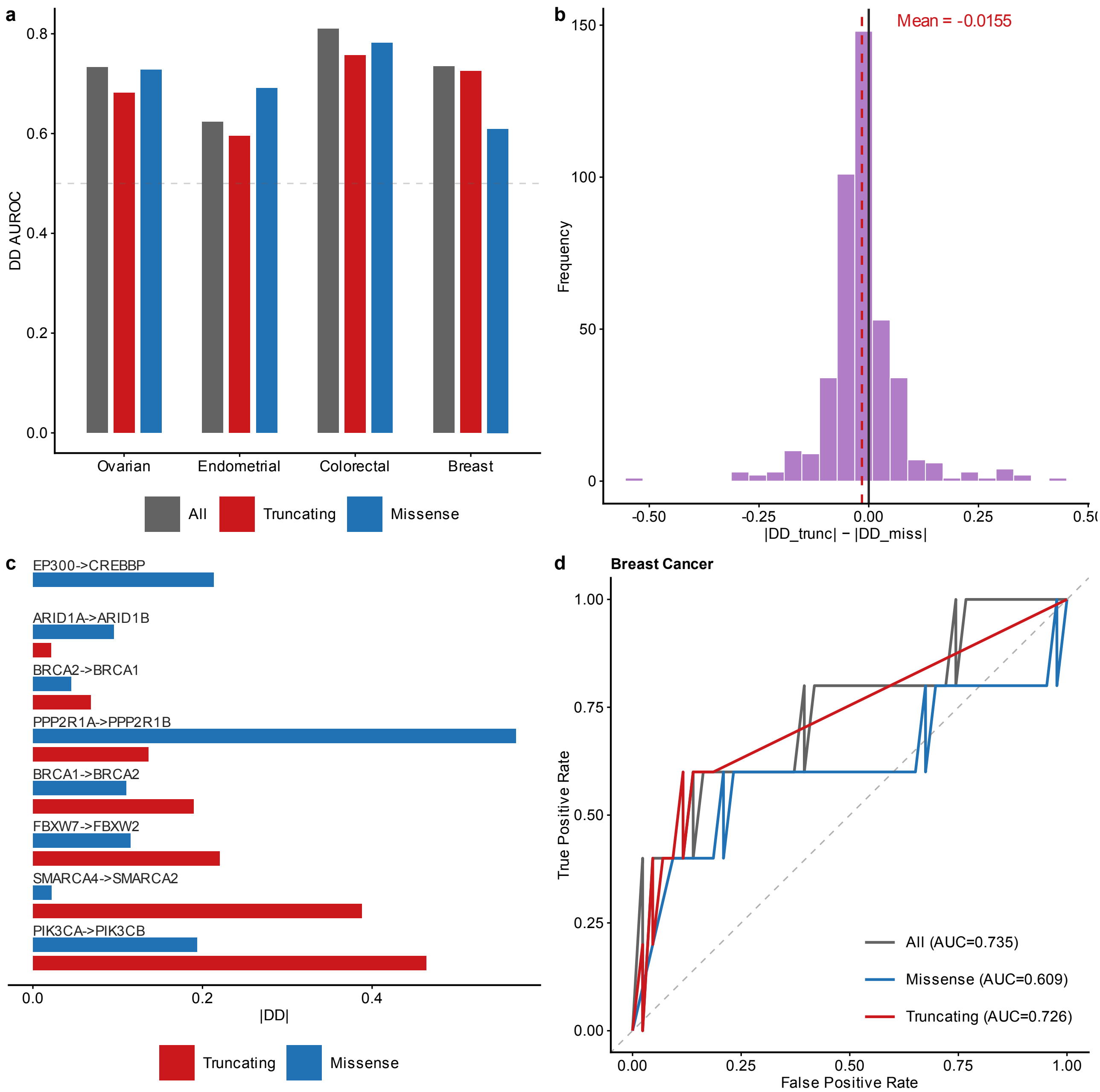

Fig. S6: Mutation-type stratification. a, DD AUROC stratified by mutation type (truncating vs. missense) across cancer types. b, Distribution of ΔDD (truncating minus missense) across all tested driver–paralog pairs; overall mean near zero (−0.015), indicating that truncating > missense is pair-specific rather than universal. c, ΔDD for known gold-standard SL pairs showing that ARID1A→ARID1B (+0.368 in Ovarian) and EP300→CREBBP (+0.314 in Colorectal) drive the trend. d, ROC curves for Breast cancer comparing truncating-only (AUROC = 0.726) vs. missense-only (0.391) and overall DD performance. Panel d is computed on the mutation-type analysis frame (48 Breast pairs); the resulting "All" AUC (0.735) therefore differs from the cross-cancer frame value (0.750; Fig. 1a), which uses a different pair universe.

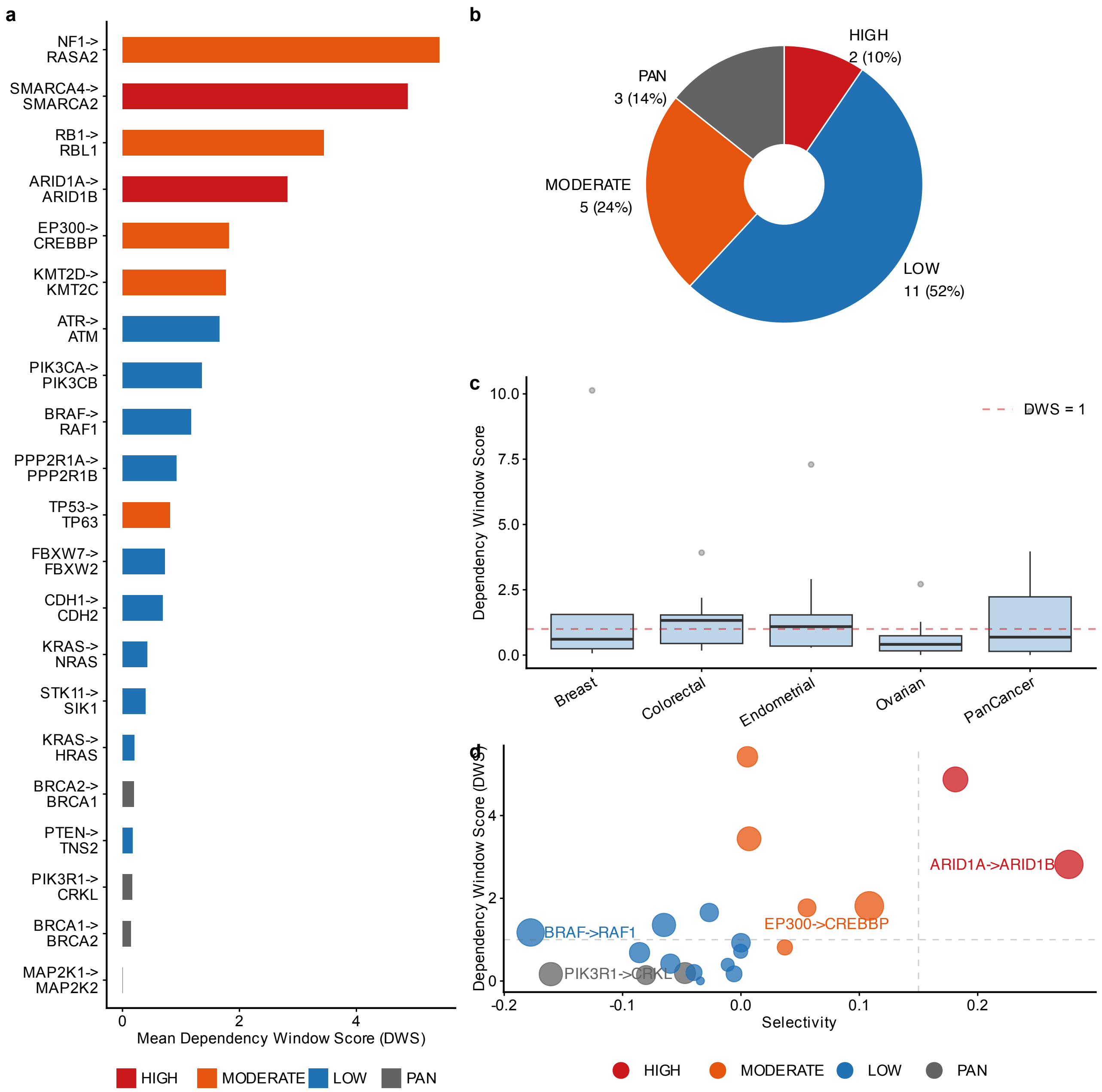

Fig. S7: Therapeutic window analysis. a, All 21 paralog pairs ranked by mean dependency window score (DWS) across up to 5 cancer contexts (Ovarian, Endometrial, Breast, Colorectal, PanCancer) on the ≥5-mutant frame. SMARCA4→SMARCA2 (4.87) and ARID1A→ARID1B (2.82) are the two HIGH_SELECTIVITY pairs; NF1→RASA2 attains a higher raw DWS through a near-zero pan-essential denominator with selectivity ≈ 0. b, In vitro selectivity tier classification: HIGH_SELECTIVITY (selectivity > 0.15 and DWS > 1.0; $n = 2$), MODERATE ($n = 5$), LOW_SELECTIVITY ($n = 11$), and PAN_ESSENTIAL ($f_{\text{pan-essential}} > 0.5$; $n = 3$). c, DWS values broken down by individual cancer context showing cross-lineage consistency for top candidates. d, Selectivity vs. DWS scatter plot; ARID1A→ARID1B and SMARCA4→SMARCA2 occupy the upper-right quadrant (high DWS and high selectivity).

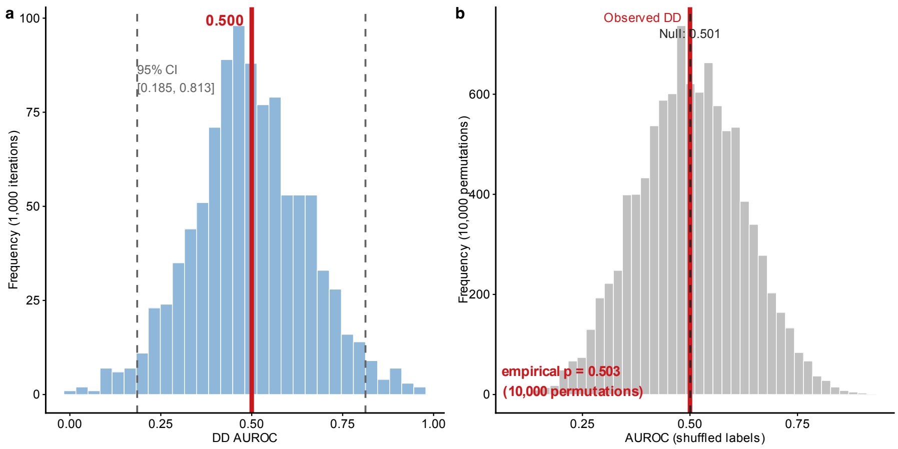

Fig. S8: Bootstrap and negative control (per-pair evaluation, 72 pairs, 6 known positives). a, Bootstrap distribution (1,000 iterations; 95% CI: 0.185–0.813). b, Null distribution from 10,000 label-shuffled permutations (empirical $p = 0.503$). The observed per-pair AUROC (0.500) does not exceed the null mean (0.501): with only 6 positive pairs the aggregated per-pair evaluation has no detectable signal, and the lineage-level evaluation (0.676, reported in the benchmark comparison) serves as the primary framework because it evaluates each driver×paralog×lineage combination separately rather than aggregating across lineages.

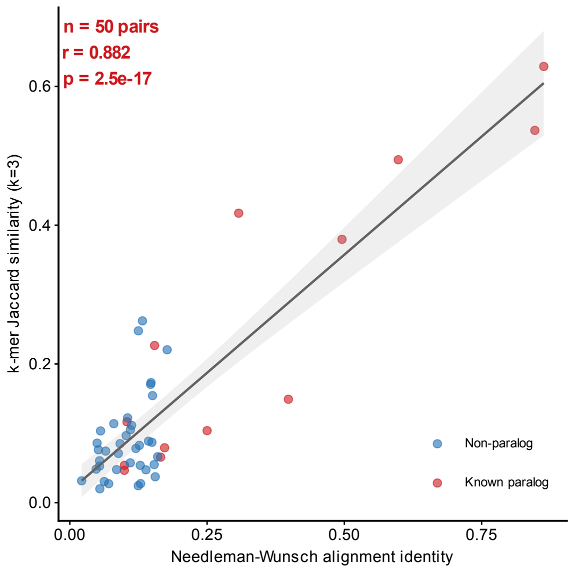

Fig. S9: k-mer Jaccard vs. Needleman-Wunsch alignment identity validation. Scatter plot of k-mer Jaccard similarity ($k = 3$) vs. global alignment identity for 50 protein pairs (13 known paralogs, 37 non-paralog controls). Red: known paralog-SL pairs; Blue: non-paralog pairs. Gray line: linear regression with 95% CI.

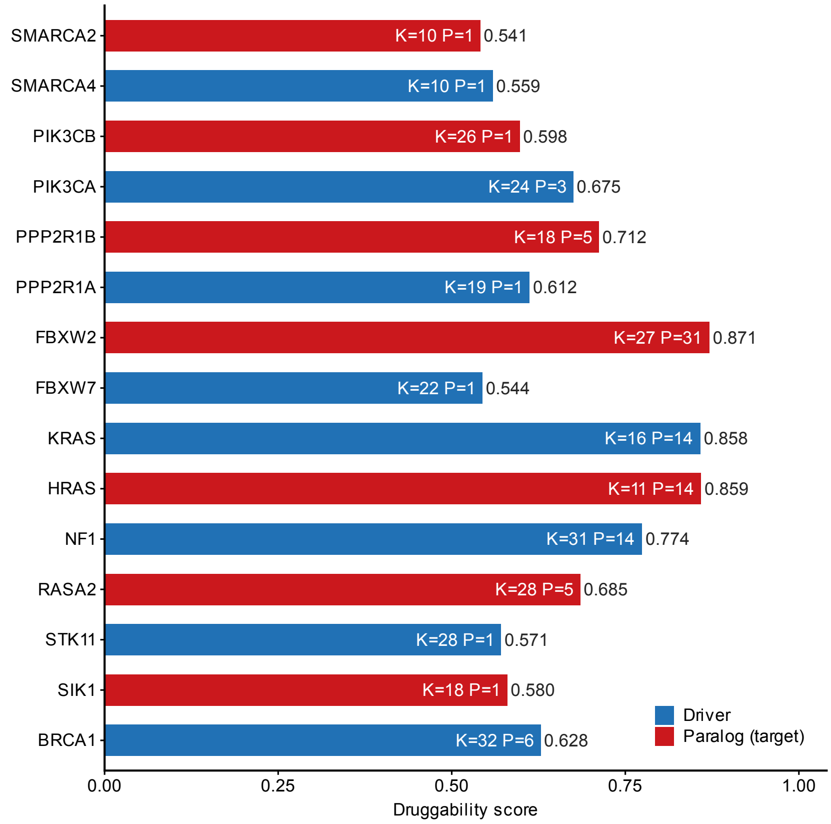

Fig. S10: Structure-based druggability assessment. Composite druggability score for 15 proteins (7 paralog targets in red, 8 driver genes in blue) based on ESMFold-predicted structures. Scores integrate structural confidence (pLDDT), hydrophobic content, lysine density (K, relevant for PROTAC conjugation), and predicted pocket count (P). Structures were limited to 400 residues per protein; ARID1A/ARID1B structures were not available from AlphaFold DB and are assessed by sequence-level features only (main text Table 2). FBXW2 (0.871) and KRAS/HRAS (0.858–0.859) show the highest scores due to well-folded compact domains.

2 Supplementary Tables

2.1 Table S1: Cell line and evaluation summary across 23 solid tumor types

Label	Full name	Cell Lines	Pairs	Known+	DD AUROC
Esophagogastric	Esophagogastric adenocarcinoma / esophageal SCC	62	50	3	0.965
SCLC	Small cell lung cancer	121	99	5	0.906
Bladder Urothelial	Bladder urothelial carcinoma	30	18	2	0.844
Colorectal	Colorectal adenocarcinoma	59	62	5	0.828
Endometrial	Endometrial carcinoma	31	62	5	0.818
Breast	Invasive breast carcinoma	50	12	2	0.750
NSCLC	Non-small cell lung cancer	95	96	5	0.741
Ovarian	Ovarian epithelial carcinoma	55	50	7	0.661
Biliary Tract	Biliary tract cancer	39	38	0	—
Melanoma	Melanoma	78	22	1	—
Glioma	Diffuse glioma	66	19	0	—
HNSCC	Head and neck squamous cell carcinoma	60	13	1	—
Pancreatic	Pancreatic adenocarcinoma	44	9	1	—
Hepatocellular	Hepatocellular carcinoma	24	4	0	—
Renal Cell	Renal cell carcinoma	26	4	0	—
Other Sarcoma	Other soft-tissue sarcoma	33	4	0	—
Cervical	Cervical carcinoma	16	3	1	—
Mesothelioma	Pleural mesothelioma	21	2	0	—
Neuroblastoma	Neuroblastoma	33	2	0	—
Osteosarcoma	Osteosarcoma	19	0	0	—
Ewing Sarcoma	Ewing sarcoma	18	0	0	—
Rhabdomyosarcoma	Rhabdomyosarcoma	12	0	0	—
Thyroid	Thyroid carcinoma	10	0	0	—

Table S1: Cell line counts and DD AUROC across all 23 analyzed solid tumor types (primary $\geq$5-per-group frame). Top: AUROC > 0.7 (7 types). Middle: AUROC ≤ 0.7 (1 type). Bottom: insufficient known positives for AUROC (15 types). Known+: number of gold-standard positive pairs evaluable in each lineage. "—": AUROC not computable ($<$2 known positives). Pairs: driver$\times$paralog combinations passing the $\geq$5 mutant / $\geq$5 wild-type filter. On the relaxed $\geq$3-per-group sensitivity frame, four additional lineages become evaluable (Biliary Tract 0.990, Pancreatic 0.949, Melanoma 0.617, Cervical 0.500). Full name: standardized cancer-type name corresponding to each short label; SCC, squamous cell carcinoma.

2.2 Table S2: Complete paralog-SL pair analysis results

The full analysis results are provided as a tab-separated file: `TableS2_FullResults.tsv` (110 rows $\times$ 17 columns, representing all driver$\times$paralog$\times$cancer-type combinations passing the minimum sample-size filter of $\geq$5 mutant and $\geq$5 wild-type cell lines). Effect sizes (Cohen's d, Hedges' g) and per-stratum Welch t-test p-values for the same 110 associations are provided as `TableS8_EffectSizes.tsv` (Supplementary Table S8).

Data dictionary:

- `driver_gene`: Driver gene with damaging mutation.
- `paralog_gene`: Candidate paralog tested for conditional dependency.
- `cancer_type`: Solid tumor lineage (DepMap OncotreePrimaryDisease).
- `pcs`: Paralog Compensation Score (DD $\times$ necessity).
- `delta_expression`: ΔExpression between MUT and WT groups.
- `necessity`: Mean gene-effect score of the paralog (more negative = more essential).

- `dependency_dd`: Delta Dependency (signed; positive = greater dependency in MUT).
- `composite_score`: Weighted composite of PCS, DD, and ΔExpression.
- `novelty`: "Known" if in gold-standard set (Table S3), else "Novel".
- `q_value`: Benjamini-Hochberg adjusted p-value within each driver gene.
- `mutation_frequency`: Fraction of cell lines with driver mutation in this lineage.
- `n_mut`, `n_wt`: Number of mutant and wild-type cell lines.
- `is_known_paralog_sl`: Boolean; TRUE if pair is in gold-standard set.

2.3 Table S3: Gold-standard paralog-SL pairs with evidence provenance and tier assignment

Tier	Driver	Paralog	Assay (perturbation)	Model system	Validated direction	Dual?	Indep.	Direct SL	Inclusion	Key ref
A	AKT1	AKT2	Combinatorial CRISPR digenic KO	Cancer cell lines	AKT1→AKT2	Yes	Yes	Yes	Primary	Najm 2018
A	CDK4	CDK6	Digenic KO (pgPEN library)	Cancer cell lines	CDK4→CDK6	Yes	Yes	Yes	Primary	Parrish 2021
A	MAP2K1	MAP2K2	Digenic KO (pgPEN library)	Cancer cell lines	MAP2K1→MAP2K2	Yes	Yes	Yes	Primary	Parrish 2021
B	SMARCA4	SMARCA2	CRISPR KO conditioned on natural SMARCA4 mutation	SMARCA4-mutant cancer lines	SMARCA4→SMARCA2	No	Yes	Conditional	Primary	Hoffman 2014
B	ARID1A	ARID1B	shRNA knockdown conditioned on natural ARID1A mutation	ARID1A-mutant cancer lines	ARID1A→ARID1B	No	Yes	Conditional	Primary	Helming 2014
C	EP300	CREBBP	p300 degradation / CRISPR in CREBBP-deficient lines	CREBBP-mutant cancer lines	Reciprocal only (CREBBP→EP300)	No	Yes	Reciprocal	Secondary	Ogiwara 2016; Nie 2021
C	PIK3CA	PIK3CB	shRNA / PI3K inhibitor in PTEN-deficient lines	PTEN-deficient cancer lines	PTEN→PIK3CB only	No	Yes	No (other driver)	Secondary	Wee 2008
C	CCNE1	CCNE2	Mouse developmental double knockout	Mouse embryo	CCNE1↔CCNE2 redundancy	Yes (mouse)	Yes	No (redundancy)	Secondary	Geng 2003
C	FBXW7	FBXW2	DepMap computational analysis	700+ cell lines	FBXW7→FBXW2	No	No	No (computational)	Secondary	DepMap
C	PPP2R1A	PPP2R1B	DepMap computational analysis	700+ cell lines	PPP2R1A→PPP2R1B	No	No	No (computational)	Secondary	DepMap
comp.	BRCA1	BRCA2	PARP inhibition / genetic screen (functional analogs)	BRCA-mutant cancer lines	BRCA1/2→PARP axis	No	Yes	Functional analog	Comparator	Bryant 2005
comp.	STK11	SIK1	Mouse genetics, LKB1–SIK axis (partial homolog)	Mouse NSCLC models	STK11→SIK1/3 axis	No	Yes	Pathway axis	Comparator	Hollstein 2019

Table S3: Twelve curated pairs with evidence provenance and tier assignment after verification of the primary citations. **Tier A** (3 pairs): direct genetic synthetic-lethal evidence from dual-gene perturbation (combinatorial/digenic CRISPR knockout). **Tier B** (2 pairs): natural-genotype conditional dependency — the paralog is a demonstrated selective dependency in driver-mutant cells — from single-gene perturbation with functional validation. The Tier A ∪ Tier B set (5 pairs) constitutes the primary external benchmark. **Tier C** (5 pairs): indirect evidence only (reciprocal-direction-only, other-driver, developmental-redundancy, or DepMap-derived); excluded from the primary benchmark and reported separately. **Comparators** (2 pairs): mechanistic reference pairs that are not sequence paralogs (functional analog or pathway axis); used as specificity references only. **Dual?**: whether the evidence comes from simultaneous dual-gene perturbation. **Indep.**: evidence independent of DepMap CRISPR data. **Direct SL**: whether the evidence directly demonstrates synthetic lethality in the direction scored in this study ("Conditional" = genotype-conditional dependency; "Reciprocal" = direct evidence for the reverse direction only). **Inclusion**: Primary = primary external benchmark; Secondary = reported separately; Comparator = specificity reference. All twelve pairs are HGNC gene-group sequence paralogs except the two comparators; validated directions refer to the driver→paralog direction scored here.

2.4 External combinatorial-CRISPR gold standards (in4mer)

Esmaeili Anvar et al. (2024) meta-analyzed five combinatorial CRISPR paralog screens (Dede 2020, Gonatopoulos-Pournatzis 2020, Parrish 2021, Thompson 2021, Ito 2021) and defined 13 candidate cross-study gold-standard paralog synthetic lethals (paralog score > 0.25, called a hit in more than one study): CNOT7–CNOT8, PITPNA–PITPNB, TIA1–TIAL1, SAR1A–SAR1B, PTP4A1–PTP4A2, GSK3A–GSK3B, CSNK2A1–CSNK2A2, CSNK1D–CSNK1E, MAPK1–MAPK3, ARFGEF1–ARFGEF2, HDAC1–HDAC2, ASF1A–ASF1B, and SLC25A28–SLC25A37. We assessed whether these pairs could serve as an external benchmark for DD. Only one of the 13 pairs (MAPK1–MAPK3) involves a gene in our 40-gene driver panel, and MAPK1 carries qualifying oncogenic hotspot mutations in zero DepMap 26Q1 cell lines (Table S7), so no in4mer gold-standard pair is evaluable in our driver-mutation-conditioned framework. This reflects a genuine division of scope rather than missing data: the in4mer gold standards measure background-independent paralog buffering under dual knockout, whereas DD quantifies dependency shifts conditioned on naturally occurring driver mutations. We therefore report the in4mer set as a conceptual cross-reference rather than a benchmark, and we note that only 8 of the 13 pairs replicated as hits in at least three of the four in4mer screens (Chou et al. 2025).

2.5 Table S4: Pan-cancer summary statistics

Cancer Type	Cell Lines	Pairs	DD AUROC
Esophagogastric	62	50	0.965
SCLC	121	99	0.906
Bladder Urothelial	30	18	0.844
Colorectal	59	62	0.828
Endometrial	31	62	0.818
Breast	50	12	0.750
NSCLC	95	96	0.741
Ovarian	55	50	0.661

Table S4: DD AUROC across all 8 evaluable solid tumor types on the primary ≥5-per-group frame (sorted by AUROC). Horizontal line separates AUROC > 0.7 (top, 7 lineages) from AUROC ≤ 0.7 (bottom). Full lineage names are given in Table S1; the ≥3-per-group sensitivity frame yields 12 evaluable lineages (9 above 0.7).

2.6 Table S5: 40-gene driver panel

Gene	Cancer Types	Source	Category
TP53	Pan-cancer	TCGA+COSMIC	TSG
KRAS	Lung, CRC, Pancreatic	TCGA+COSMIC	Oncogene
PIK3CA	Breast, Endometrial, Cervical	TCGA+COSMIC	Oncogene
PTEN	Endometrial, Ovarian, Breast	TCGA+COSMIC	TSG
ARID1A	Endometrial, Ovarian, CRC	TCGA+COSMIC	TSG
BRCA1	Ovarian, Breast	TCGA+COSMIC	TSG
BRCA2	Ovarian, Breast	TCGA+COSMIC	TSG
BRAF	Melanoma, CRC, Thyroid	TCGA+COSMIC	Oncogene
NF1	Glioma, Breast, Lung	TCGA+COSMIC	TSG
RB1	SCLC, Breast, Bladder	TCGA+COSMIC	TSG

Table S5: First 10 of 40 driver genes (complete list in TableS5_DriverGenes.tsv). Selection criteria: recurrently mutated in ≥1 TCGA pan-cancer study or COSMIC Cancer Gene Census, with ≥5 mutant cell lines in at least one DepMap lineage.

2.7 Table S6: All 21 paralog pairs ranked by dependency window score

Driver	Paralog	|DD|	DWS	Select.	Class	Type
NF1	RASA2	0.101	5.421	0.005	MOD	Seq
SMARCA4	SMARCA2	0.186	4.873	0.181	HS	Seq
RB1	RBL1	0.164	3.439	0.007	MOD	Seq
ARID1A	ARID1B	0.270	2.819	0.277	HS	Seq
EP300	CREBBP	0.279	1.815	0.108	MOD	Seq
KMT2D	KMT2C	0.064	1.771	0.056	MOD	Seq
ATR	ATM	0.070	1.656	−0.027	LS	Seq
PIK3CA	PIK3CB	0.152	1.356	−0.065	LS	Seq
BRAF	RAF1	0.245	1.174	−0.178	LS	Seq
PPP2R1A	PPP2R1B	0.074	0.925	0.000	LS	Seq
TP53	TP63	0.035	0.813	0.037	MOD	Seq
FBXW7	FBXW2	0.025	0.720	0.000	LS	Seq
CDH1	CDH2	0.098	0.687	−0.086	LS	Seq
KRAS	NRAS	0.083	0.420	−0.060	LS	Seq
STK11	SIK1	0.017	0.392	−0.011	LS	Part
KRAS	HRAS	0.043	0.206	−0.040	LS	Seq
BRCA2	BRCA1	0.111	0.193	−0.047	PE	Func
PTEN	TNS2	0.042	0.174	−0.006	LS	Seq
PIK3R1	CRKL	0.152	0.167	−0.161	PE	Seq
BRCA1	BRCA2	0.079	0.144	−0.080	PE	Func
MAP2K1	MAP2K2	0.001	0.003	−0.034	LS	Seq

Table S6: All 21 analyzed paralog pairs ranked by mean dependency window score (DWS) on the ≥5-mutant frame. |DD|: mean absolute Delta Dependency across cancer contexts. Select.: mean selectivity (fraction mutant-essential minus fraction wild-type-essential). Class: HS=HIGH_SELECTIVITY, MOD=MODERATE, LS=LOW_SELECTIVITY, PE=PAN_ESSENTIAL. Type: Seq=sequence paralog, Part=partial homology, Func=functional analog. Three pairs evaluable under the previous frame (AKT1→AKT2, CCNE1→CCNE2, CDK4→CDK6) no longer meet the ≥5-mutant threshold in any context and are not listed.

2.8 Table S7: Gene-class-specific driver-mutation rules and variant classification

Gene	Class	Rule	Old	New	% ret.
ALK	ONC	Hotspot	118	12	10.2
APC	TSG	LikelyLoF	217	108	49.8
ARID1A	TSG	LikelyLoF	240	173	72.1
ATM	TSG	LikelyLoF	165	59	35.8
ATR	TSG	LikelyLoF	125	37	29.6
BRAF	ONC	Hotspot	212	162	76.4
BRCA1	TSG	LikelyLoF	101	34	33.7
BRCA2	TSG	LikelyLoF	159	68	42.8
CCNE1	ONC	Hotspot	22	0	0.0
CDH1	TSG	LikelyLoF	87	34	39.1
CDK4	ONC	Hotspot	27	8	29.6
CTNNB1	ONC	Hotspot	85	45	52.9
EGFR	ONC	Hotspot	110	19	17.3
EP300	TSG	LikelyLoF	206	107	51.9
ERBB2	ONC	Hotspot	70	20	28.6
FBXW7	TSG	LikelyLoF	116	81	69.8
GATA3	TSG	LikelyLoF	52	16	30.8
KEAP1	TSG	LikelyLoF	99	31	31.3
KMT2D	TSG	LikelyLoF	356	185	52.0
KRAS	ONC	Hotspot	254	233	91.7
MAP2K1	ONC	Hotspot	36	14	38.9
MAP3K1	TSG	LikelyLoF	74	14	18.9
MAPK1	ONC	Hotspot	13	0	0.0
MET	ONC	Hotspot	66	1	1.5
NF1	TSG	LikelyLoF	215	122	56.7
PIK3CA	ONC	Hotspot	182	153	84.1
PIK3R1	TSG	LikelyLoF	79	46	58.2
PPP2R1A	TSG	LikelyLoF	53	13	24.5
PTEN	TSG	LikelyLoF	199	176	88.4
RB1	TSG	LikelyLoF	185	150	81.1
SMAD4	TSG	LikelyLoF	119	80	67.2
STK11	TSG	LikelyLoF	97	66	68.0
TP53	TSG	LikelyLoF	1171	1140	97.4

Table S7: Driver-mutation definitions by gene class (DepMap 26Q1, default-entry profiles). TSG: mutant requires `LikelyLoF`; ONC: mutant requires `Hotspot`. Old: cell lines carrying any non-silent variant (pre-revision permissive definition); New: cell lines qualifying under the class-specific rule; % ret.: 100 $\times$ New/Old. The top VariantInfo terms per gene are at `output/driver_mutation_rules.csv`. CCNE1 and MAPK1 are amplification-driven oncogenes with no qualifying hotspot mutations.